

Eventual Coherence: Preserving Causal Validity in Replicated Systems

Solving the Problem of Divergent Intent in Shared State Networks

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Abstract

The delete-update conflict, where one device deletes a resource while another concurrently updates it, is a fundamental underdetermined challenge in multi-device synchronization. Existing approaches treat it as a synchronization problem rather than a semantics problem and so sacrifice at least one of three properties: availability through coordination, causal validity through last-write-wins semantics, or storage boundedness through conflict-free replicated data types. We argue that this tradeoff is unnecessary and arises from a category error: treating topology operations (CREATE, DELETE) and content operations (UPDATE) as equivalent writes subject to uniform reconciliation semantics. We introduce eventual coherence, a synchronization model grounded in the categorical distinction between these operation classes. Control plane operations establish resource topology and are reconciled at synchronization boundaries. Data plane operations modify resource content and propagate in near-real-time. Reconciliation is governed by two principles. First, local deletion finality: a device that has deleted a resource permanently ignores all subsequent updates to it. Second, server-level upsert semantics: a server receiving a data plane update for a deleted resource restores it, ensuring devices with active causal participation continue receiving updates. Together these principles resolve the delete-update conflict without coordination, without tombstones, and without violating causal validity for any participating device. The formal contribution of this paper is the rejection of the classical consistency hierarchy as the sovereign framework for evaluating human-facing distributed systems. We argue that optimizing for uniform cross-replica agreement represents a framework error when devices act as causal proxies for human agents. Eventual coherence decouples system evaluation into two orthogonal dimensions: inter-replica agreement and local historical validity. The protocol is implementable over standard HTTP without vector clocks, operational transformation, or physical clock synchronization assumptions.

1. Introduction

Distributed systems that maintain replicated state across multiple devices face a persistent challenge: how to reconcile operations when devices independently modify shared resources [1, 2]. This problem becomes particularly acute when one device deletes a resource while another concurrently updates it. Consider a conversation management system where Device A deletes a conversation, while Device B, operating without knowledge of this deletion, continues adding messages to it. When these devices synchronize, the system must decide: should the deletion propagate, discarding Device B's work, or should the update propagate, implicitly reversing the deletion?

Neither outcome is correct. Device A made a deliberate choice to delete. Device B made a deliberate choice to continue working. A system that erases either choice in the name of convergence has not achieved correctness. It has imposed an arbitrary resolution on a situation that does not require one.

Traditional approaches to this problem fall into three categories, each sacrificing at least one property the other two preserve. Coordination-based systems achieve consistency by requiring devices to obtain locks [3] or reach consensus [4, 5, 6] before performing operations, but this approach sacrifices availability when devices cannot communicate [7], precisely the conditions under which multi-device systems are most valuable [8, 9]. Last-write-wins (LWW) strategies avoid coordination by imposing total ordering based on timestamps [10, 11], but this discards causal validity: a deletion performed after an update may be overridden by that update due to clock skew [13], or an update performed after a deletion may be discarded when the deletion carries a later timestamp, losing work performed in good faith. Conflict-free replicated data types [14, 15, 16] preserve all operations through carefully designed merge semantics, but make true deletion difficult, requiring either tombstones that accumulate without bound [21, 76] or monotonic state growth [20] that prevents genuine removal.

All of these approaches introduce substantial implementation complexity, forcing application developers to reason about specialized data structures, merge semantics, and long-term state management concerns that arise primarily from the synchronization mechanism itself rather than the underlying application domain.

We observe that this problem arises from a category error that runs through all three approaches: treating all operations as equivalent writes subject to uniform reconciliation semantics. Resource creation and deletion establish the topology of the system, what resources exist at all. Resource updates modify the content of existing resources without changing topology. These are different kinds of operation having different causal properties [12, 22], different timing requirements, and different correct reconciliation behaviors. Handling them uniformly produces systems that are simultaneously harder to build and less faithful to user intent.

This paper introduces eventual coherence, a synchronization model built on the explicit categorical separation of these operation classes. Control plane operations (CREATE, DELETE) establish resource topology. They are reconciled at synchronization boundaries, defined points at which devices exchange control plane state with the server. Data plane operations (UPDATE) modify resource content. They propagate in near-real-time throughout normal device operation. This separation is not a performance optimization. It reflects a genuine difference in the nature of the operations and their role in establishing system invariants.

Reconciliation in eventual coherence is governed by two principles. The first is local deletion finality: a device that has deleted a resource permanently ignores all subsequent data plane updates targeting that resource. The deletion is a terminal state for that device's relationship with the resource. The second is server-level upsert semantics: when a server receives a data plane update for a resource currently marked as deleted, it restores the resource. This ensures that devices with active causal participation in the resource continue to receive updates from other participants.

These two principles together resolve the delete-update conflict. The deleting device maintains its deletion. Devices that were actively using the resource continue to do so. Third-party devices receive the most recent updates. No coordination is required. No tombstones accumulate. No causal validity is violated for any participating device.

The formal contribution of this paper is the introduction of eventual coherence, a correctness criterion that operates outside the classical distributed consistency hierarchy. The traditional hierarchy is fundamentally an agreement hierarchy; it assumes that a distributed system's proximity to correctness is proportional to its total and fine-grained consensus. This assumption holds for systems modeling a single source of objective reality, such as financial ledgers. It fails catastrophically when replicas are extensions

of independent human actors. A system that erases an explicit local action to force cross-replica agreement has not achieved correctness; it has achieved conformity by destroying information.

Eventual coherence rejects the assumption that agreement is the sole proxy for correctness, treating inter-replica agreement and historical validity as independent, orthogonal dimensions of a broader correctness space. This criterion is not universally appropriate. Systems where replicas must agree on a single, shared authoritative state, such as financial ledgers or inventory management, rightly prioritize consistency. Eventual coherence targets systems where devices represent independent human agents operating in distinct causal contexts, and where forcing convergence means erasing legitimate work.

This approach provides several advantages over existing methods. Unlike coordination-based systems [4, 5, 31], it operates correctly without requiring devices to communicate before performing operations. Unlike last-write-wins [10, 33], it preserves causal validity: updates performed in the context of a resource's existence are never discarded due to timestamp ordering, and deletions are never overridden by remote data plane operations on the deleting device. Unlike conflict-free replicated data types [14, 15], it requires no operational transformation, supports true deletion, and does not require unbounded storage growth. Most significantly, it achieves these properties without vector clocks, without distributed consensus, and with a protocol simple enough to implement over standard HTTP [36].

We make the following contributions:

- A formal definition of eventual coherence as a correctness criterion grounded in Lamport clock ordering, establishing its precise relationship to existing consistency models and proving it is incomparable to eventual consistency rather than merely weaker.
- A synchronization protocol that achieves eventual coherence through two principles, local deletion finality and server-level upsert semantics, requiring no physical clock synchronization, no vector clocks, and no operational transformation.
- Proofs that the protocol guarantees per-device causal validity, local deletion finality, and convergence to causally valid states within the bounds of Lamport clock ordering.
- A proof-of-concept implementation in Polyglot, a multi-device conversation management system, demonstrating practical realizability using standard web technologies, with a full reference implementation in progress.

The remainder of this paper is organized as follows. Section 2 presents our system model and formal definitions. Section 3 describes the eventual coherence protocol. Section 4 proves correctness properties. Section 5 describes the proof-of-concept implementation. Section 6 evaluates performance. Section 7 surveys related work, and Section 8 concludes.

2. System Model and Definitions

We model a distributed system consisting of a finite set of continuously connected devices $D = \{d_1, d_2, \dots, d_n\}$ that replicate a shared set of resources R . Each resource r in R has a unique identifier and represents a container for mutable content (in our implementation, a conversation consisting of messages). Devices perform operations on resources and synchronize with a server S through a disciplined message passing model described below. This model follows the client-server architecture common in mobile and web applications [8, 41] rather than the peer-to-peer model of some replicated systems [42, 43].

A central premise of this model is that continuous connectivity does not imply, nor require, uniform synchronization of all operation classes. Devices deliberately propagate different classes of operations at different frequencies. This is not a concession to network unreliability. It is an architectural principle reflecting the categorical difference between the operations themselves.

2.1 Synchronization Boundaries

We introduce synchronization boundaries as a first-class concept in the model. A synchronization boundary is a point at which a device performs control plane reconciliation with S. The definition of a synchronization boundary is left to the implementor: it may correspond to application launch, page load, explicit user action (such as initialization), or any other meaningful transition in device state. Between synchronization boundaries, a device operates within a local synchronization epoch, during which it processes data plane operations in near-real-time but does not pull control plane state from S.

Definition 1 (Synchronization Epoch). *A synchronization epoch for device d_i is the interval between two consecutive synchronization boundaries. Within a synchronization epoch, d_i processes data plane operations in near-real-time and accumulates control plane operations for reconciliation at the next synchronization boundary.*

2.2 Lamport Clocks

All causal ordering in the eventual coherence model is established via Lamport clocks [13]. No assumptions are made about physical clock synchronization across devices.

Definition 2 (Lamport Clock). *Each device d_i maintains a local Lamport counter L_i , initialized to 0, obeying the following rules:*

- *On each local operation: $L_i := L_i + 1$*
- *On sending a message: the current value of L_i is included in the message.*
- *On receiving a message carrying Lamport value L_{msg} : $L_i := \max(L_i, L_{msg}) + 1$*

The server S maintains a global Lamport counter L_s updated by the same rule on every message received.

Definition 3 (Lamport Timestamp). *Each operation o carries a Lamport timestamp $L(o) = (L_i, d_i)$ where d_i is the issuing device, used for deterministic tie-breaking. The total order on Lamport timestamps is: (L_1, d_1) precedes (L_2, d_2) if $L_1 < L_2$, or $L_1 = L_2$ and d_1 precedes d_2 lexicographically. This total order is consistent with the happens-before relation [13].*

Physical timestamps may be maintained for human-readable display purposes but are not load-bearing in the protocol. All ordering decisions use Lamport timestamps exclusively.

2.3 Operations

We define two categorically distinct classes of operations. This distinction is the architectural foundation of eventual coherence and reflects a genuine difference in the nature of the operations, not merely a difference in propagation timing.

Definition 4 (Control Plane Operations). A control plane operation is a function that modifies the topology of the resource set R . Control plane operations are pushed to S immediately upon execution and pulled from S only at synchronization boundaries. We consider two control plane operations:

- $CREATE(r)$: Add resource r to R , with $r \notin R$ before the operation.
- $DELETE(r)$: Mark resource $r \in R$ as deleted on the issuing device.

Definition 5 (Data Plane Operations). A data plane operation is a function that modifies the content of an existing resource without changing resource topology. Data plane operations propagate to S in near-real-time throughout a synchronization epoch.

- $UPDATE(r, \delta)$: Apply modification δ to resource $r \in R$.

Each operation o carries a Lamport timestamp $L(o)$ as defined in Definition 3, a device identifier $o.d$, and a resource identifier $o.r$. This operational model differs from state-based CRDTs [14, 20], which transmit entire states, and operation-based CRDTs [15, 49], which require causal delivery [50]. Our model requires neither complete state transmission nor causal broadcast [51].

2.4 Causal History

Definition 6 (Local History). The local history H_i of device d_i is the sequence of operations performed by d_i totally ordered by Lamport timestamp $L(o)$.

Definition 7 (Causal Participation). Device d_i causally participates in resource r with respect to a deletion record $\tau_{\text{delete}} = (l_d, d_d)$ if there exists an operation $o \in H_i$ such that $o.r = r$ and the Lamport timestamp of o , written $\tau_o = (l_o, d_o)$, strictly dominates τ_{delete} lexicographically: $l_o > l_d$, or $(l_o = l_d \text{ and } d_o > d_d)$. We write $\tau_o > \tau_{\text{delete}}$ to denote this relation.

Causal participation determines how a device responds to control plane operations at synchronization boundaries. A device that has modified a resource with a Lamport timestamp that strictly dominates the deletion record has a causal stake in the resource's continued existence. This notion tracks participation rather than complete causal history, which is both sufficient for our purposes and significantly simpler to implement than vector clock approaches [34, 35, 52].

2.5 System Assumptions

We assume an asynchronous network model [53] where message delays are unbounded but finite. Devices may crash and recover [55]. We assume crash-recovery failures only, with no Byzantine behavior [56]. S is assumed to be available and crash-recoverable with persistent storage [57].

S serves two functions. First, it maintains persistent current state for all resources, enabling devices to synchronize data plane state at any time. Second, it maintains a history of control plane events sufficient for devices to reconcile at synchronization boundaries. S broadcasts data plane updates to all connected devices in near-real-time. S does not maintain per-device state and has no knowledge of which devices have performed which local operations. S is not a source of truth for individual device state. It is a shared communication and persistence layer.

Devices maintain local replicas in persistent storage [58] and perform all operations locally without coordination [59, 60]. Data plane operations are pushed to S immediately. Control plane operations are pushed to S immediately but pulled from S only at synchronization boundaries. We make no assumption about operation ordering across devices, as in the asynchronous model of distributed computing [53, 61]. This model is weaker than systems requiring quorum reads and writes [62, 63] but stronger than systems with no server coordination [42, 64]. It reflects the practical reality of modern mobile and web applications where a shared server provides a convergence point without requiring real-time coordination for all operation classes [8, 65].

A note on scope: The model assumes data plane operations on a given resource originate from a single author operating across multiple devices. Concurrent authorship, with distinct authors independently targeting the same resource within the same synchronization epoch, is outside the model's scope. (Under concurrent authorship, Lamport-ordered last-write-wins at the resource level discards one author's operations, violating Property 6.) Multi-author data plane semantics are left to a subsequent model.

2.6 Consistency Properties and Orthogonality

Classical distributed consistency models operate on a foundational assumption: that a system's proximity to correctness is directly proportional to its inter-replica agreement. We argue this is a framework error when replicas serve as distinct causal proxies for human intent. To evaluate such systems, we decouple correctness into two orthogonal dimensions: the Agreement Axis and the Coherence Axis. Let $\sigma(d_i, t)$ represent the materialized state of a resource set on device d_i at logical time t .

Definition 8 (The Agreement Axis). *A system satisfies the Agreement Axis if for any two devices $d_i, d_j \in D$, their materialized states asymptotically converge under a terminal execution trace:*

$$\lim_{t \rightarrow \infty} \sigma(d_i, t) = \sigma(d_j, t)$$

The Agreement Axis as defined here corresponds to the standard formulation of eventual consistency [10]: absent continued updates, all replicas converge to an identical state.

Definition 9 (The Coherence Axis / Historical Validity). *A system satisfies the Coherence Axis for device d_i if, for every locally committed operation $o \in H_i$ that is causally admissible, the operational mutations induced by o are deterministically preserved in $\sigma(d_i, t)$ regardless of any concurrent remote operations $o' \in H_j$ ($j \neq i$). An operation o targeting resource r is causally admissible if no deletion record exists for r , or if τ_o strictly dominates the deletion record τ_{delete} for r lexicographically ($\tau_o > \tau_{\text{delete}}$). Operations for which $\tau_o \not\triangleright \tau_{\text{delete}}$ lie outside the causal horizon and may be superseded by the deletion; such supersession is not a coherence violation.*

Definition 10 (Eventual Coherence). *A system provides eventual coherence if, for any resource r , when all operations on r cease and all pending synchronization boundaries have been crossed, each device's replica of r converges to a state that is causally valid given that device's local history, as ordered by the Lamport happens-before relation.*

Proposition 1 (Orthogonality of Frameworks). *The Agreement Axis and the Coherence Axis are mutually orthogonal correctness criteria. A system may satisfy the Agreement Axis while violating the Coherence Axis on one or more devices, or conversely, satisfy the Coherence Axis on all devices while maintaining permanent state divergence ($\lim_{t \rightarrow \infty} \sigma(d_i, t) \neq \sigma(d_j, t)$).*

We defer the formal proof of the Orthogonality Proposition to Section 4.5.

2.7 The Reconciliation Problem

Given this model, we state the problem eventual coherence solves.

Problem Statement (The Reconciliation Problem). *Design a synchronization protocol such that:*

1. *Devices can perform data plane operations continuously within a synchronization epoch without coordination.*
2. *Control plane operations propagate to S immediately upon execution and to all devices at synchronization boundaries.*
3. *Data plane operations synchronize in near-real-time throughout each synchronization epoch.*
4. *After a synchronization boundary is crossed, each device's state is causally valid with respect to the Lamport happens-before relation.*
5. *A device that has deleted a resource is never required to restore it due to remote data plane operations.*
6. *No device loses data plane operations it performed within a synchronization epoch, provided those operations do not conflict with concurrent data plane operations from a distinct author targeting the same resource.*
7. *Deletion metadata does not accumulate without bound; once all devices have crossed a synchronization boundary since a deletion and made their participation decision, the corresponding metadata is eligible for garbage collection.*

Traditional approaches violate at least one of these requirements. Coordination-based systems [4, 31] violate (1) due to the CAP theorem [7, 75]. Last-write-wins systems [10, 33] violate either (5) or (6) depending on timestamp ordering. CRDT systems [14, 15] make true deletion difficult, violating (7) through tombstone accumulation [21, 76] or monotonic state growth [20]. Systems using operational transformation [17, 18] achieve (6) but at significant implementation complexity [19]. Our protocol satisfies all seven properties through explicit plane separation, Lamport clock ordering, local deletion finality, and server-level upsert semantics.

3. The Protocol

The eventual coherence protocol consists of four components: local operation execution, real-time data plane synchronization, real-time control plane push, and control plane reconciliation at synchronization boundaries. We present each component in turn.

3.1 Local Operation Execution

All operations execute locally without coordination [59, 77]. When device d_i performs an operation o on resource r , it immediately applies the operation to its local replica, increments its Lamport counter, and records o in its local history H_i . This follows the optimistic replication paradigm [2, 78].

For control plane operations:

```

CREATE(r):
    Li := Li + 1
    r.id <- generate_unique_id()
    r.lamport <- (Li, di)
    r.created_by <- di
    r.content <- empty
    local_storage.put(r)

DELETE(r):
    Li := Li + 1
    r.is_deleted <- true
    r.deleted_at_lamport <- (Li, di)
    r.deleted_by <- di
    local_storage.put(r)

```

DELETE does not remove the resource from local storage. It marks the resource as deleted and records the Lamport timestamp of the deletion for use during subsequent control plane reconciliation.

For data plane operations:

```

UPDATE(r, delta):
    if r.is_deleted then
        discard silently // Invariant 5: Local Deletion Finality
        return
    Li := Li + 1
    r.content <- apply(r.content, delta)
    r.lamport <- (Li, di)
    r.modified_by <- di
    local_storage.put(r)

```

If a device has locally marked a resource as deleted, all subsequent data plane operations targeting that resource are silently discarded. The deletion is a terminal state for that device’s relationship with the resource. This is Invariant 5 (Local Deletion Finality), stated formally in Section 3.5.

3.2 Real-Time Data Plane Synchronization

When a device performs an UPDATE operation, it immediately propagates the update to S:

```

on UPDATE(r, delta):
    apply_locally(r, delta)
    send_to_server(UPDATE_MESSAGE, r, r.clock)

```

S receives updates, advances its global Lamport counter, and merges using Lamport-ordered last-write-wins semantics:


```

on receive UPDATE_MESSAGE(r, clock) from device di:
  Ls := max(Ls, clock.lamport) + 1
  r_server <- server_storage.get(r.id)

  if r_server is null then
    server_storage.put(r)
    broadcast_to_connected_devices(r)
    return

  if r_server.is_deleted then
    // Upsert: restore resource so other devices receive this update.
    // Deletion record is preserved.
    // NB: v1 correct under single-author assumption only.
    r_server.is_deleted <- false

  if compareLamport(r.clock, r_server.clock) > 0 then
    r_server.messages <- r.messages
    r_server.clock <- r.clock
    server_storage.put(r_server)
    broadcast_to_connected_devices(r_server)

```

The upsert behavior is the mechanism by which third-party devices receive updates from participants who are still actively using a resource. When S receives a data plane update for a resource currently marked as deleted, it restores the resource and broadcasts the update to all connected devices. S preserves the deletion record (`deleted_at_lamport`) after the upsert. This record is required for control plane reconciliation at synchronization boundaries and is never cleared by data plane operations. Devices that have locally deleted a resource receive restoration broadcasts but discard them under Invariant 5. The broadcast reaches and correctly updates third-party devices.

Crucially, the protocol coordinator must remain entirely decoupled from structural payload heuristics. An update payload containing an empty message array (`messages: []`)—as seen during initial conversation initialization—must be processed as a valid state mutation rather than being short-circuited into a deletion pathway. Treating structural emptiness as an implicit cleanup operation violates protocol separation, resulting in deletion calls on resources that have not yet established a persistence baseline.

The upsert mechanism presupposes that the receiving device already possesses some local record of the target resource. This need not hold: a device may create a resource and immediately update it within a single synchronization epoch, and the resulting broadcast may reach a third-party device that has not yet crossed a boundary and therefore has no topological knowledge of the resource at all. The receive handler at every non-originating device must account for this case explicitly:

```

on receive UPDATE_BROADCAST(r) at device di (not the originating device):
  Li := max(Li, r.clock.lamport) + 1
  // clock advances regardless content app
  r_local <- local_storage.get(r.id)

  if r_local is null then
    signal_boundary_available() // Invariant 6: Data Plane Non-Origination
    return

  if r_local.is_deleted then
    discard // Invariant 5: Local Deletion Finality
    return

  if compareLamport(r.clock, r_local.clock) > 0 then
    r_local.messages <- r.messages
    r_local.clock <- r.clock
    local_storage.put(r_local)

```

A device with no local topological record of r does not apply the broadcast's content to local storage. Materializing a new local resource from a data plane event would allow an UPDATE to perform the function of a CREATE, violating the categorical separation of Definitions 4 and 5. However, the broadcast is causally real (the Lamport clock advance confirms this) and its arrival is affirmative evidence that a CREATE exists at S which the device has not yet pulled. The prohibition is therefore on content materialization, not on response: the device signals that a synchronization boundary is available.

A compliant implementation may surface this signal as a user-visible badge indicating that a boundary is ready, or may use it to trigger an immediate boundary crossing via an automated presentation layer event. Both behaviors preserve the control/data plane separation, since the data plane event is used as a routing signal rather than a content source. Topological acquisition of r , with full content, proceeds at the next synchronization boundary. S already holds the authoritative merged state; no content is lost by deferring to the boundary.

3.3 Real-Time Control Plane Push

Control plane operations are pushed to S immediately upon local execution. This is the push side of the asymmetric control plane propagation model: devices push control plane events in real-time but pull them only at synchronization boundaries.

```

on CREATE(r):
  apply_locally(r)
  send_to_server(CREATE_MESSAGE, r, r.clock)

on DELETE(r):
  apply_locally(r)
  record <- DeletionRecord(r.id, r.clock)
  send_to_server(DELETE_MESSAGE, record)

```

S receives control plane events and updates its persistent state:

```
on receive CREATE_MESSAGE(r, clock) from device di:
  Ls := max(Ls, clock.lamport) + 1
  r_server <- server_storage.get(r.id)
  If r_server is not null then return // Reject duplicates / race conditions
  server_storage.put(r)

on receive DELETE_MESSAGE(record) from device di:
  Ls := max(Ls, record.clock.lamport) + 1
  r_server <- server_storage.get(record.id)
  if r_server is null then return

  r_server.clock <- record.clock
  r_server.deleted_by <- di
  r_server.is_deleted <- true
  server_storage.put(r_server)
  server_storage.track_deletion(record)
  // Deletion is NOT broadcast to connected devices in real-time.
```

S does not broadcast deletions to connected devices in real-time. Deletions propagate to devices only at synchronization boundaries. This is the pull side of the asymmetric model. Data plane updates arriving after a deletion, from devices that have not yet crossed a synchronization boundary and learned of it, trigger upserts at S, restoring the resource and broadcasting updates to connected devices. Devices that have locally deleted the resource discard these broadcasts under Invariant 5.

This asymmetry has an important operational consequence: during the period between a deletion event and a device's next synchronization boundary, the server may oscillate between deleted and restored states for a resource as the deleting device's control plane push and other devices' data plane updates alternate. This oscillation is an inherent, expected architectural property of an optimistically replicated system where the server functions as a transport intermediary rather than a centralized convergence authority. It does not constitute a correctness violation. The oscillation resolves deterministically when all active participants cross their respective synchronization boundaries and commit their local participation decisions.

3.4 Control Plane Reconciliation at Synchronization Boundaries

When device d_i crosses a synchronization boundary, it reconciles its local state with S:

```

on synchronization_boundary():
    // Step 1: Advance Lamport clock from server tracking metadata
    Ls_current <- fetch_from_server(LAMPORT_COUNTER)
    Li := max(Li, Ls_current) + 1

    // Step 2: Fetch current server data plane and control plane states
    server_resources <- fetch_from_server(ALL_RESOURCES)
    server_deletions <- fetch_from_server(ALL_DELETION_RECORDS)

    // Step 3: Reconcile local state map against server state
    for each r_server in server_resources do
        r_local <- local_storage.get(r_server.id)
        r_del_record <- server_deletions.find(id == r_server.id)

        // Case 1: Local Deletion Finality
        if r_local exists and r_local.is_deleted then
            record <- DeletionRecord(r_local.id, r_local.clock)
            send_to_server(DELETE_MESSAGE, record)
            continue

        // Case 2: Causal Participation Detection via Clock Primitives
        if r_del_record is not null then
            participated := r_local is not null and compareLamport(r_local.clock, r_del_record.clock) >
0
            if not participated then
                local_storage.delete(r_server.id)
                local_storage.track_deletion(r_del_record)
            else
                send_to_server(UPDATE_MESSAGE, r_local)

        // Case 3: Standard Last-Write-Wins (LWW) Data Merge
        else
            if r_local is null or compareLamport(r_server.clock, r_local.clock) > 0 then
                local_storage.put(r_server)

    // Step 4: Push independent local-only resources to server
    for each r_local in local_storage.get_all() do
        if r_local.id not in server_resources and not r_local.is_deleted then
            send_to_server(CREATE_MESSAGE, r_local, r_local.clock)

    last_boundary_clock <- CoherenceClock(Li, di)

```

Three aspects of this algorithm require explanation. Case 1 is evaluated unconditionally before all other cases. A device that has locally deleted a resource never proceeds to the deletion record check or the LWW merge. The device re-informs S of its local deletion, which re-asserts the deletion at S and supersedes any upsert-driven restoration. This is Local Deletion Finality in the reconciliation context.

Case 2 implements causal participation detection via full lexicographic Lamport tuple ordering. A device's local update strictly dominates the deletion record if and only if its Lamport counter exceeds the deletion's counter, or, in the equal-counter case, its device identifier is lexicographically greater than the deleting device's identifier. If the local update does not strictly dominate, including the case where logical counters are equal and the local device identifier does not dominate, the algorithm defaults to deleted. This guarantees deterministic local convergence: concurrent operations at the same logical counter always resolve in favor of the destructive control plane operation, without requiring cross-device coordination.

The default to deleted behavior for new devices falls out naturally from Case 2. A device with no local record of a resource has no causal participation. If the resource carries a deletion record, the device accepts the deletion. A new device does not acquire a resource it never participated in.

Case 3 handles resources with no deletion record through standard Lamport-ordered last-write-wins. No participation decision is required.

3.5 Correctness Invariants

Invariant 1 (Local Causal Consistency). *Within a single synchronization epoch on device d_i , all operations respect the happens-before relation [13]. If $o1 \rightarrow o2$ on d_i , then $L(o1) < L(o2)$ in H_i . This is guaranteed by the Lamport clock rules: each operation increments L_i before executing.*

Invariant 2 (Control Plane Idempotence). *Applying the same control plane operation multiple times has the same effect as applying it once. CREATE is idempotent because creating an already-existing resource is a no-op. DELETE is idempotent because marking a deleted resource as deleted again changes nothing. This ensures that network failures and retries do not violate correctness [85, 86].*

Invariant 3 (Data Plane Convergence). *If all devices cease data plane operations and all pending data plane synchronizations complete, all devices that have not deleted a resource converge to the same content for that resource. This follows from Lamport-ordered last-write-wins: the version with the highest Lamport timestamp eventually propagates to all non-deleted replicas.*

Invariant 4 (Causal Participation Preservation). *If a device d_i executes a data plane update u on a resource R within its local history H_i , the operational intent of u is globally preserved across subsequent synchronization boundaries, provided that the device's local Lamport counter has been advanced past the deletion counter ld associated with any remote control plane deletion of R prior to executing u .*

Invariant 5 (Local Deletion Finality). *If device d_i has marked resource r as deleted, all subsequent data plane operations targeting r on d_i are silently discarded. Local deletion is a terminal state for a device's relationship with a resource. It cannot be overridden by remote data plane operations. This invariant applies both during normal operation (Section 3.1) and during control plane reconciliation (Section 3.4, Case 1). To sustain this invariant, a device that deletes r must retain a minimal local deletion record, i.e. $\{r.id, r.deletedAtLamport\}$. This record must not be removed by any data plane operation. Its presence is what allows the receive handler (Section 3.2) to distinguish a deleted resource from a resource the device has never seen. Without it, both cases produce a null result from `local_storage.get(r.id)`, and the two invariants become unenforceable simultaneously.*

The retention requirement of Invariant 5 is what makes a null result from `local_storage.get(r.id)` semantically unambiguous. With it in force, a device that returns null for some resource id is guaranteed to have never acquired that resource; deleted resources remain present in local storage with `is_deleted=true`. This guarantee is load-bearing for the case where a device may receive a real-time data plane broadcast for a resource it has no topological knowledge of, e.g. when another device creates and updates a resource within a single synchronization epoch before the receiving device has crossed a boundary. The receive handler must distinguish this case from a locally deleted resource and respond differently to each, which distinction is only available if Invariant 5's retention requirement holds.

A device may receive a real-time data plane broadcast for a resource it has no topological knowledge of, for instance when another device both creates and updates a resource within a single synchronization epoch before the receiving device has crossed a boundary. The following invariant governs this case.

Invariant 6 (Data Plane Non-Origination). A device that receives a real-time data plane broadcast for a resource r for which it holds no local record, neither active nor deleted, does not apply the broadcast content to local storage. The device's Lamport counter advances per Definition 2's unconditional receive rule. The broadcast's arrival constitutes affirmative evidence that a control plane CREATE for r exists at S which the device has not yet pulled; the device therefore signals that a synchronization boundary is available. Topological acquisition of r , including its content as of the broadcast, is deferred to the device's next synchronization boundary, where Case 3 of the reconciliation algorithm merges the server's current state for r via standard Lamport-ordered last-write-wins. The invariant prohibits content materialization from data plane events; it does not prohibit using such events as boundary availability signals. Compliant implementations may surface this signal as a user-visible notification or use it to trigger an immediate boundary crossing.

3.6 Server Metadata Immutability

To guarantee that the coordination environment can accurately preserve execution history for evaluation by reconciling devices, the server must enforce absolute metadata durability for deletion records.

Invariant 7 (Deletion Record Timestamp Preservation). *Let DR be the server-side deletion record for a deleted resource R , containing the definitive logical deletion timestamp $\tau_{deletedAt}$. From the moment of initial control plane registration until final metadata garbage collection, $\tau_{deletedAt}$ must be strictly immutable. The coordination server is explicitly prohibited from overwriting, shifting, or truncating $\tau_{deletedAt}$ in response to incoming data plane updates.*

3.7 Garbage Collection

Client-side deletion records are subject to a symmetric purge condition. A device may remove its local DeletionRecord from its underlying data store when it crosses a synchronization boundary and the target resource ID is absent from the server's resource list. Absence at a synchronization boundary serves as the observable signal that S has completed server-side garbage collection for the resource, which guarantees that no future data-plane updates or broadcasts for that resource ID will be issued.

The deletion record's disambiguating function is exhausted at that point. A device must not purge a local deletion record on any other basis; elapsed time, storage pressure, or application-layer signals are

not safe purge triggers, as none of them establish that a resource ID will never reappear in a future broadcast.

```
/* Deletion records can be garbage collected once all devices have crossed a
synchronization boundary and no device has retained the resource */

on server_garbage_collect():
    for each record in server_storage.get_all_deletion_records() do
        if all_devices_crossed_boundary_since(record.clock) then
            if no_device_retained(record.id) then
                server_storage.delete_resource(record.id)
                server_storage.purge_deletion_record(record.id)
```

This formulation requires S to track synchronization boundary crossings per device, which is minimal per-device state. V2 of the protocol eliminates this requirement by having devices explicitly signal completion of synchronization boundary processing, removing server-side tracking entirely.

Device lifecycle presents an open problem for garbage collection. Devices that are permanently decommissioned or lost will block garbage collection indefinitely if S awaits their synchronization boundary crossing. Characterizing the interaction between device lifecycle and garbage collection is left to future work.

3.8 Complexity Analysis

Communication Complexity. Data plane updates require $O(1)$ messages per operation (device to server, server broadcast to connected devices). Control plane push requires $O(1)$ messages per control plane operation. Control plane reconciliation at a synchronization boundary requires $O(|R|)$ messages, where $|R|$ is the number of resources. Only Lamport timestamps and deletion records are exchanged during reconciliation, not complete resource state.

Storage Complexity. Each device stores $O(|R_{\text{active}}|)$ resources, where R_{active} is the subset of R not deleted from that device's perspective. S stores $O(|R|)$ resources plus $O(|R_{\text{deleted}}|)$ deletion records pending garbage collection. Unlike operation-based CRDTs [15, 49] that must store complete operation logs [88], eventual coherence requires only current state plus deletion records.

Time Complexity. Local operations complete in $O(1)$ time. Reconciliation at a synchronization boundary requires $O(|R|)$ comparisons. This is optimal: the device must examine each resource to determine its participation status.

4. Correctness Proofs

We prove that the eventual coherence protocol satisfies its correctness properties. Proofs follow the style of Lamport [13, 69] and Lynch [32], establishing properties through invariant reasoning grounded in the Lamport happens-before relation.

4.1 Per-Device Causal Validity

We begin by establishing the primary correctness property of eventual coherence. We do not claim that the protocol satisfies classical causal consistency, which Ahamad et al. [39] define as requiring all devices to observe causally related operations in the same order globally. The protocol explicitly permits permanent divergence between devices with incompatible divergent causal histories. We claim instead the following.

Theorem 1 (Per-Device Causal Validity). *For any device d_i , the eventual coherence protocol guarantees that d_i 's local state is causally valid with respect to its local history H_i at all times,*

Proof. We must show that for any operation o in H_i , the causal dependencies of o are respected in d_i 's local state.

Case 1: o is a data plane operation (UPDATE). Data plane operations execute locally and immediately. By Invariant 1, operations within a synchronization epoch are totally ordered by Lamport timestamps. If o causally depends on a prior operation o' in H_i , then $L(o') < L(o)$ by the Lamport happens-before relation [13]. Since d_i applies operations in Lamport timestamp order, o' is applied before o . The causal dependency is respected.

Case 2: o is a control plane CREATE operation. CREATE establishes a resource that subsequent operations may depend on. Since CREATE executes locally and increments L_i before recording r in local storage, any operation o' in H_i that depends on r satisfies $L(o') > L(\text{CREATE}(r))$. The causal dependency is respected within H_i .

Case 3: o is a control plane DELETE operation. DELETE marks r as deleted locally. By Invariant 5 (Local Deletion Finality), no subsequent operation in H_i may target r . Therefore no operation in H_i causally depends on r existing after $L(\text{DELETE}(r))$. d_i 's local state correctly reflects that r does not exist after the deletion.

Case 4: d_i crosses a synchronization boundary and discovers a deletion record for resource r at Lamport timestamp L_d .

Sub-case 4a: r_{local} is null, or $\tau_{local} \not\prec \tau_{delete}$ (r_{local} 's lexicographic timestamp does not strictly dominate the deletion record) and d_i has no operation in H_i whose Lamport tuple strictly post-dates the deletion. By Definition 7, d_i did not causally participate in r after the deletion. Accepting the deletion introduces no causal violation: d_i has no operations in H_i that causally depend on r existing after the deletion record.

Sub-case 4b: $\tau_{local} > \tau_{delete}$. d_i has at least one operation o in H_i with $o.r = r$ whose lexicographic Lamport tuple strictly dominates the deletion record. By Definition 7, d_i causally participated in r after the deletion.

By Invariant 4 (Causal Participation Preservation), d_i retains r and o 's operations on r after the deletion form a causally consistent sequence under the Lamport happens-before relation: each operation depends on the state of r as d_i observed it within its synchronization epoch.

Since d_i did not observe the deletion before performing these operations (if d_i had, Invariant 5 would have prevented further updates), these operations are causally valid. Note that the retained operations may be concurrent with the deletion in the happens-before sense: neither d_i 's updates nor the deletion causally preceded the other. The Lamport ordering establishes a consistent total order between concurrent events without implying causal precedence [13]. This is the correct treatment of concurrency under the Lamport model.

In all cases, d_i 's local state respects the causal dependencies in H_i . QED.

4.2 Convergence to Causally Valid States

Theorem 2 (Convergence). *When all operations cease and all pending synchronization boundaries have been crossed, each device converges to a state that is causally valid given its local history.*

Proof. We show that the reconciliation algorithm terminates in a stable state satisfying Definition 10. Let $T_{quiesce}$ be the time when all operations cease. Let T_{sync} be the time when all pending synchronization boundaries have been crossed. We claim that at time $T > T_{sync}$, every device's state is causally valid.

Step 1: Reconciliation terminates. Each reconciliation processes $O(|R|)$ resources with constant-time comparisons. Since $|R|$ is finite and no new operations occur after $T_{quiesce}$, reconciliation completes in finite time. Let T_{recon} be the time at which all devices complete reconciliation. T_{recon} is finite.

Step 2: The resulting state is causally valid for each device. Consider device d_i at time $T > T_{recon}$. For each resource r , exactly one of the following holds.

(a) d_i has locally deleted r . By Theorem 4 (Local Deletion Stability, proved below), d_i 's deletion is stable across all synchronization boundaries. d_i 's state is causally valid by Case 3 of Theorem 1.

(b) R carries no deletion record and d_i holds the latest version by Lamport-ordered last-write-wins. d_i 's state is causally valid by Cases 1 and 2 of Theorem 1. Case (b) additionally covers devices that received but did not apply real-time broadcasts for resources outside their topological knowledge under Invariant 6; receipt triggered a boundary availability signal rather than content materialization. Such devices acquire the resource through the same Case 3 mechanism at their next synchronization boundary, with no loss of content, which remains durably held at S .

(c) R carries a deletion record and d_i 's local Lamport tuple for r strictly dominates the deletion record lexicographically. d_i causally participated after the deletion and retains r . Thus, d_i 's state is causally valid by Case 4b of Theorem 1.

(d) R carries a deletion record and d_i has no local record of r , or d_i 's Lamport tuple for r does not strictly dominate the deletion record. d_i accepts the deletion and thus d_i 's state is causally valid by Case 4a of Theorem 1.

In all cases, d_i 's final state respects all causal dependencies in H_i , satisfying Definition 10. Cases (a) and (c) may produce permanently divergent states between devices. This divergence is the correct outcome under Definition 10, not a failure. Eventual coherence does not require cross-device convergence to identical state. It requires only that each device converge to a state that is causally valid for that device. QED.

4.3 Update Preservation

Theorem 3 (Update Preservation). *Let τ_o be the logical timestamp of a data plane operation o committed locally on device d_i . Upon crossing a subsequent synchronization boundary, o is deterministically preserved in d_i 's materialized state unless there exists a control plane deletion record τ_{delete} such that $\tau_o \not\prec \tau_{delete}$.*

Proof. At the synchronization boundary, d_i reconciles its local history against the server state. We evaluate by operation class:

Case 1: o is a CREATE(r) operation. The new resource topology is pushed to the server during reconciliation Step 4. Because resource identifiers are globally unique, r is cleanly established or verified at the server, preserving o 's structural intent.

Case 2: o is an $UPDATE(r, \delta)$ operation. If r carries no server-side deletion record, o is preserved trivially via Lamport-ordered last-write-wins propagation. If r carries a deletion record τ_{delete} , the client applies Case 2 reconciliation (§3.4): if $\tau_o > \tau_{delete}$, d_i establishes active causal participation, retaining r and preserving o 's modification; if $\tau_o \not> \tau_{delete}$, o fails to strictly dominate the deletion horizon and is safely superseded.

Case 3: o is a $DELETE(r)$ operation. Under Case 1 of the reconciliation algorithm, d_i unconditionally re-asserts its local deletion to the server. By Theorem 4 (Local Deletion Stability), d_i maintains r as deleted, satisfying and preserving its destructive intent permanently. QED.

4.4 Local Deletion Stability

Theorem 4 (Local Deletion Stability). *If device d_i has marked resource r as deleted, then for all subsequent synchronization boundary crossings, d_i maintains r as deleted in its local state.*

Proof. At each synchronization boundary crossing, the reconciliation algorithm evaluates Case 1 first and unconditionally: if r_{local} exists and $r_{local}.is_deleted$ is true, the algorithm sends `DELETE_MESSAGE` to S and proceeds to the next resource without evaluating Cases 2 or 3. No path through the reconciliation algorithm sets $r_{local}.is_deleted$ to false for a locally deleted resource. No incoming data plane update can modify a locally deleted resource (Invariant 5). Therefore once d_i has marked r as deleted, d_i 's local state has r as deleted at every subsequent synchronization boundary crossing. QED.

Corollary 1. *Local deletion finality is permanent absent an explicit control plane `CREATE` operation issued by an administrative authority with a Lamport timestamp higher than the existing deletion record. Such an operation, received by d_i at a synchronization boundary, would appear as a new resource in S 's state without a deletion record post-dating the `CREATE`, allowing d_i to acquire the resource through Case 3 reconciliation. The issuance and authorization of such operations is outside the scope of the synchronization protocol.*

The synchronization protocol supports, but does not implement, a second administrative operation: a full resynchronization that issues a `CREATE` for every resource currently retained by any device but deleted on the requesting device, restoring the requesting device's complete view in one pass. This operation enters the protocol as a batch of explicit control plane `CREATE` events with Lamport timestamps exceeding the corresponding deletion records; the protocol's only obligation is to preserve the state needed to construct them, which Invariant 7 (Deletion Record Timestamp Preservation, Section 3.6) guarantees. The issuance, authorization, and sequencing of both administrative operations are application-layer concerns outside the scope of the synchronization protocol.

4.5 Proof of the Orthogonality Proposition

We now prove the Orthogonality Proposition formulated in Section 2.6. To demonstrate that the Agreement Axis and the Coherence Axis are mutually orthogonal, we must establish two distinct conditions: (1) that a system can achieve perfect inter-replica agreement while comprehensively violating historical validity, and (2) that a system can guarantee perfect historical validity while permitting permanent state divergence.

Lemma 1 (Agreement Without Coherence). *Let A be a synchronization system whose correctness criterion requires global convergence to a single terminal state. Consider concurrent operations $DELETE(r)$ and $UPDATE(r, \delta)$. Any deterministic reconciliation rule in A that satisfies the Agreement Axis by producing a unique, unified terminal state ($\sigma(d_1) = \sigma(d_2)$) must discard the structural effect of at least one committed operation. If the system converges to a deleted state, the committed update is erased; if it converges to an active state, the terminal deletion is reversed. In either execution path, system A violates the Coherence Axis for at least one device, establishing that satisfaction of the Agreement Axis does not imply satisfaction of the Coherence Axis.*

Lemma 2 (Coherence Without Agreement). *Let B be a synchronization system operating under the eventual coherence protocol. Consider the identical concurrent scenario: d_1 issues $DELETE(r)$ and d_2 issues $UPDATE(r, \delta)$. Suppose d_2 's local Lamport counter has been advanced past $\tau_{delete} = \langle ld, dd \rangle$ prior to issuing $UPDATE(r, \delta)$. Upon crossing a synchronization boundary, under Case 1 of the reconciliation algorithm, d_1 unconditionally maintains its local deletion state ($\sigma(d_1) = \emptyset$). Concurrently, under Case 2, because our setup assumption guarantees that $\tau_{local} > \tau_{delete}$, d_2 retains the resource and materializes its operational update ($\sigma(d_2) = \{r \mapsto \delta\}$). As $t \rightarrow \infty$, the states remain permanently diverged: $\sigma(d_1) \neq \sigma(d_2)$. System B violates the Agreement Axis, yet it fully satisfies the Coherence Axis for all participating nodes, as the operational mutations of both local agents are preserved.*

By Lemma 1 and Lemma 2, Proposition 1 holds: satisfaction of one axis implies nothing about the satisfaction of the other. The Agreement Axis and the Coherence Axis are formally independent, orthogonal dimensions of distributed state evaluation.

Corollary 2 (Incomparability with Eventual Consistency). *Eventual coherence and eventual consistency are incomparable: eventual consistency does not imply eventual coherence, and eventual coherence does not imply eventual consistency.*

Proof. The Agreement Axis (Definition 8) is the formal expression of eventual consistency within this model. By Lemma 1, a system satisfying the Agreement Axis (and therefore eventual consistency) may fully violate the Coherence Axis; eventual consistency does not imply eventual coherence. By Lemma 2, a system satisfying the Coherence Axis on all devices may maintain permanent state divergence, violating the Agreement Axis and therefore eventual consistency; eventual coherence does not imply eventual consistency. Since neither property implies the other, the two are incomparable. QED.

4.6 Liveness

Theorem 5 (Liveness). *If all messages are eventually delivered and all devices eventually cross synchronization boundaries, the system reaches a stable state in finite time.*

Proof. Assume eventual message delivery and eventual synchronization boundary crossings. We show that the system reaches a fixed point. Let $T_{quiesce}$ be the time when all operations cease. After $T_{quiesce}$, no new operations enter any device's local history. Each subsequent synchronization boundary crossing makes deterministic decisions based on Lamport timestamps and local history. Since no new operations occur after $T_{quiesce}$, these inputs do not change. By Invariant 2 (Control Plane Idempotence),

repeated synchronization boundary crossings after $T_{quiesce}$ produce identical decisions. By Theorem 4 (Local Deletion Stability), deletion decisions made at synchronization boundaries are stable. By Invariant 3 (Data Plane Convergence), data plane state converges to the version with the highest Lamport timestamp among non-deleted replicas. Therefore there exists a time $T > T_{quiesce}$ after which no device’s state changes at any synchronization boundary crossing. The system has reached a fixed point. This fixed point may include permanently divergent states between devices. A device that has deleted a resource and a device that has retained it will remain divergent indefinitely. This is a stable and correct terminal state, not a failure mode. The liveness property guarantees termination of the reconciliation process, not convergence to identical state. QED.

5. Conclusion

We have presented eventual coherence, a synchronization model for replicated state in multi-device human-facing systems. The model rests on two foundational observations. First, resource existence and resource modification are categorically different operations with different causal properties, different timing requirements, and different correct reconciliation behaviors. Handling them uniformly, as most synchronization protocols do, produces systems that are simultaneously harder to build and less faithful to user intent. Second, for systems where devices represent independent human agents operating in distinct causal contexts, convergence to identical state is not the correct correctness criterion. Per-device causal validity is.

The formal contribution of this paper is the definition of eventual coherence as a correctness criterion that operates outside the classical distributed consistency hierarchy. Rather than measuring correctness by agreement, we decouple system evaluation into two orthogonal dimensions: inter-replica agreement (the Agreement Axis) and local historical validity (the Coherence Axis) and prove that neither implies the other. Corollary 2 establishes the direct consequence: eventual coherence and eventual consistency are incomparable, with neither property subsuming the other. We have proved that the protocol guarantees per-device causal validity (Theorem 1), convergence to causally valid states (Theorem 2), update preservation (Theorem 3), local deletion stability (Theorem 4), and liveness (Theorem 5). We have not claimed classical causal consistency, which requires global ordering properties incompatible with the intentional divergence eventual coherence permits. We have claimed something more appropriate for the systems we target: each device converges to a state that is correct given its own operational history, with causal ordering established through the happens-before relation rather than physical clock synchronization.

The practical contribution is a protocol built on two principles. Local deletion finality makes a device’s deletion of a resource a permanent terminal state impervious to remote data plane operations. Server-level upsert semantics ensures that devices with active causal participation propagate their updates to third-party devices even in the presence of concurrent deletions. Together these principles resolve the delete-update conflict without coordination, without tombstones, without operational transformation, and without physical clock synchronization assumptions. The protocol is implementable over standard HTTP.

Eventual coherence is not the right model for every distributed system. Systems where replicas must agree on a single authoritative state, such as financial ledgers or inventory management, rightly prioritize consistency. The contribution is scoped specifically to systems where devices represent people with independent agency, where concurrent activity is the norm, and where forcing agreement means erasing legitimate work.

Future work falls into four areas. The forthcoming full reference implementation of Polyglot will provide a rigorous empirical foundation for the performance claims of Section 6 and a platform for comparative evaluation against optimized alternative implementations. The relationship between synchronization boundary frequency and control plane staleness warrants formal characterization: how often must boundaries be crossed to bound the divergence between devices, and what are the practical implications for implementors? The device lifecycle problem, specifically the interaction between device decommissioning and garbage collection of deletion records, requires treatment beyond what the proof-of-concept addresses. Finally, extending eventual coherence to multi-server topologies while preserving its correctness properties is an open problem: the current model assumes a single coordination server, and relaxing this assumption will require careful treatment of the server-level upsert mechanism and Lamport counter synchronization across servers.

The deeper point this paper makes is about what distributed correctness means for systems that serve people rather than machines. The classical consistency hierarchy asks whether replicas agree. Eventual coherence asks whether each replica’s state is valid for the agent it serves — and demonstrates that these are orthogonal questions, with neither implying the other. For an increasingly important class of systems, the second question is the right one to be asking.

Appendix I: Related Work

Eventual coherence sits at the intersection of several research traditions. We survey each in turn, positioning our contribution precisely against the closest related work.

I.1 Limits of Convergent Approaches

The Lineage of Optimistic Replication and the Limits of Convergent Consensus

The foundations of modern collaborative software rest upon early optimistic replication frameworks designed to maximize availability in weakly connected environments. Chief among these is Bayou [1], which pioneered the use of per-replica operation logs, tentative executions, and background anti-entropy protocols to resolve concurrent writes without synchronous coordination. To ensure that temporary divergence does not degrade into permanent state fracturing, Bayou relies on a designated primary site to assign a definitive, monotonically increasing commit sequence number to every write operation. Replicas continuously roll back tentative local states and replay operations in accordance with this global sequence, ensuring that all nodes eventually arrive at an identical ledger.

Eventual coherence differs fundamentally from these early optimistic replication models in its definition of distributed correctness. Bayou assumes that all replicas must ultimately converge to an identical state determined by a primary commit site, treating concurrent divergence as an expected transient state during anti-entropy and reconciliation to be eliminated through global serialization. In contrast, eventual coherence treats concurrent deletion and modification not as a conflict requiring resolution, but as potentially intentional bifurcation. Where Bayou treats permanent divergence as a correctness failure, eventual coherence preserves the validity of both deletion and continued participation within their respective causal histories, elevating certain forms of permanent divergence to the correct outcome for human-facing systems.

This conceptual divergence manifests in the underlying mechanics of the reconciliation plane. Because Bayou’s correctness criterion is identity, it enforces a total order that inevitably creates an

operational winner and loser when confronting concurrent modifications to the same entity; a user’s local, tentative deletion can be summarily overturned and resurrected by a subsequent commit sequence from the primary site. Eventual coherence replaces this centralized serialization with asymmetric reconciliation rules executed at synchronization boundaries via Lamport timestamps. By partitioning operations into discrete planes and enforcing Local Deletion Finality, our protocol guarantees that a device’s choice to terminate its participation with a resource is structurally irreversible, allowing divergent data-plane timelines to coexist without violating causal validity.

The asymmetric control plane propagation model this enables, immediate push but boundary-only pull, has been observed in mobile computing systems [28, 54] and client-server synchronization protocols [8, 65], but has not previously been formalized as a correctness model with precise reconciliation semantics and provable guarantees. The observation that devices push control plane events immediately while deferring their pull to synchronization boundaries is, to our knowledge, novel as a formal protocol element with defined correctness guarantees.

I.2 Eventual Consistency and Large-Scale Systems

The practical systems literature on eventual consistency is anchored by Dynamo [10], which demonstrated that last-write-wins timestamp ordering and vector clock-based reconciliation could support high availability at scale while tolerating network partitions. Cassandra [70] and Riak [71] extended this work, with Riak subsequently adding CRDT support to address the semantic limitations of pure last-write-wins reconciliation. These systems treat all operations as equivalent writes resolved through timestamp ordering or causal dependency tracking. This uniform treatment is appropriate when the application domain does not distinguish between operations that establish resource existence and operations that modify resource content. Eventual coherence observes that for an important class of systems this distinction is fundamental, and that treating topology and content operations uniformly produces reconciliation semantics that are simultaneously harder to implement correctly and less faithful to user intent.

Dynamo’s use of physical timestamps for last-write-wins is vulnerable to clock skew [13]. Eventual coherence uses Lamport timestamps for all ordering decisions, properly grounding the protocol in the happens-before relation and eliminating dependence on physical clock synchronization. For data plane operations, eventual coherence also uses last-write-wins, but under Lamport ordering rather than physical time, inheriting Dynamo’s simplicity while correcting its formal grounding.

I.3 Conflict-Free Replicated Data Types and Operational Transformation

Conflict-Free Replicated Data Types (CRDTs) [14, 15, 16] offer a principled framework for coordination-free eventual consistency by leveraging mathematical properties that guarantee convergence regardless of network routing, delay, or operation reordering. State-based CRDTs [14, 20] evaluate state mutations as monotonic joins over a bounded join-semilattice, while operation-based CRDTs [15, 49] achieve the same end state by delivering commutative operations over a causally ordered transport [50]. The structural limitation of CRDTs when applied to the multi-device delete-update problem is not merely an engineering concern regarding state growth; it is a fundamental semantic mismatch dictated by their underlying algebraic invariants. By definition, a state-based CRDT requires that for any two concurrent states x and y , the merge operation must compute the least upper bound (the join) within a semilattice: $x \sqcup y$. This algebraic formulation enforces a critical system invariant: if a network achieves quiescence, replicas must converge to an identical, uniform state ($x \sqcup y = y \sqcup x$). Consequently, a CRDT lattice is structurally incapable of representing a permanent causal split.

When confronted with a concurrent delete-update pair, a CRDT must enforce a deterministic, global policy to maintain lattice validity, either by prioritizing deletion globally (as in standard Observed-Remove Sets [15]) or by preserving the updated value globally within a multi-value container (an MV-CRDT). Neither approach can satisfy the requirements of human-facing multi-device synchronization. If the system defaults to a “delete wins” policy, active updates performed in good faith by a user on a separate device are permanently erased. Conversely, if an MV-CRDT is employed to preserve both the deleted state and the concurrent update, the protocol merely defers the conflict to the application layer, forcing every device on the network to track, store, and eventually resolve the multi-value set. This introduces a subtle but severe violation of Local Deletion Finality (Invariant 5): a device that has explicitly deleted a resource is structurally barred from forgetting it. The deleting device is forced to maintain a persistent causal relationship with the dead resource, processing remote data-plane updates it has already opted out of, simply to prevent the global lattice from fracturing.

Furthermore, while modern CRDT literature proposes various optimizations for deletion record garbage collection and delta-state minimization [21, 76], these mitigation strategies invariably break the pure, zero-coordination deployment model. They rely on environmental assumptions such as physical clock synchronization, bounded network delivery latencies, or periodic stable-vector consensus protocols to safely purge stale metadata without risking state regression. Under the eventual coherence model, we reject both the coordination tax of garbage collection and the semantic tyranny of the join-semilattice. By decoupling the control and data planes, eventual coherence permits a radical departure from CRDT invariants: it allows the deleting device to enter a stable, terminal state where it completely purges and ignores the resource, while allowing the retaining device to preserve its data plane mutations. The system achieves local causal validity without forcing a uniform least upper bound across divergent human intents.

Operational transformation (OT) [17, 18] similarly preserves concurrent updates but requires the maintenance of complex operation logs and the evaluation of transform functions whose mathematical correctness is notoriously fragile [19]. Systems utilizing OT at scale, such as Jupiter [89], introduce significant architectural and computational complexity to resolve concurrency. Eventual coherence bypasses the algorithmic overhead of OT entirely. Rather than attempting to transform or synthesize conflicting concurrent operations, our protocol partitions operations into planes with asymmetric reconciliation rules, reducing the complex delete-update paradox to a localized, deterministic Lamport timestamp comparison executed at synchronization boundaries.

I.4 Control Plane and Data Plane Separation

While the separation of control and data planes is an established primitive in networking and distributed storage, existing architectures leverage this isolation to optimize or enforce uniform convergence. We are unaware of a prior synchronization model that utilizes plane separation to formalize intentional, permanent divergence as a valid correctness criterion for human-facing multi-device state. Software-defined networking [44] separates the control plane, which establishes network topology and routing policy, from the data plane, which forwards packets according to that policy. These planes operate at different timescales with different consistency requirements: control plane updates are relatively infrequent and may tolerate higher latency, while data plane operations must be fast and local. This is structurally identical to the distinction eventual coherence formalizes between CREATE and DELETE operations on the one hand and UPDATE operations on the other.

Burns et al. [90] describe a related distinction between desired state, declared through control plane operations and reconciled by controllers, and runtime state, which reflects the current operational behavior of the system. Version control systems such as Git maintain a similar separation between refs,

which establish the topology of the commit graph, and file contents, which are modified through commits. Eventual coherence formalizes this separation as a synchronization model for replicated state. The contribution is not the separation itself, which has clear antecedents in these systems traditions, but its formalization as a correctness criterion with precise reconciliation semantics, Lamport clock grounding, and provable invariants including Local Deletion Finality and Causal Participation Preservation.

I.5 The CAP Theorem and Availability

Brewer’s CAP theorem [7], formalized by Gilbert and Lynch [75], establishes that a distributed system cannot simultaneously provide consistency, availability, and partition tolerance. Systems that require coordination for conflict resolution, including two-phase commit [4] and Paxos-based consensus [5, 6, 31], sacrifice availability under partition in exchange for consistency guarantees. Eventual coherence accepts this tradeoff explicitly, choosing availability. The contribution goes beyond simply accepting the tradeoff: the paper argues that for the class of systems it targets, consistency is not the correct objective even when achievable. The CAP theorem frames the choice as a reluctant concession to the realities of distributed systems. Eventual coherence reframes it as a principled design decision grounded in the semantics of the operations and the nature of the agents performing them. Forcing convergence between a device that has deliberately deleted a resource and a device that has retained it through active causal participation does not improve correctness. It imposes an arbitrary resolution on a situation that does not require one.

Appendix II. Proof of Concept and Forthcoming Implementation

Polyglot is an experimental proof-of-concept system that demonstrates the practical realizability of the eventual coherence protocol using standard web technologies. This implementation uses a multi-device AI conversation management system which allows a user to maintain chat histories across multiple devices and AI providers. It is not a production system and is not presented as one. Its purpose is to demonstrate that the protocol can be realized without specialized distributed systems infrastructure, and to surface engineering considerations that a full reference implementation will need to address. A full reference implementation is in progress.

II.1 System Architecture

Polyglot consists of three components. Client devices run a React-based web application that stores conversation state in IndexedDB, a browser-native key-value store. IndexedDB provides persistent storage with ACID transactions within a single device, making it suitable for maintaining local replicas. Each browser context constitutes a device in the sense of Definition 4: it maintains an independent local history, its own Lamport counter, and crosses synchronization boundaries independently. The server is a Node.js HTTP service that maintains persistent current state for all resources and broadcasts data plane updates to connected devices in near-real-time. As established in Section 2.5, *S* is a shared communication and persistence layer. It does not maintain per-device state and has no knowledge of which devices have performed which local operations. Protocol implementation spans both client and server, realizing the algorithms from Section 3. The client side consists of approximately 800 lines of TypeScript across three modules: `indexedDbStorage.ts` (local replica management and Lamport counter persistence), `conversationStateManager.ts` (operation execution and synchronization boundary logic), and `syncService.ts` (HTTP communication).

Polyglot consists of three components. Client devices run a React-based web application that stores conversation state via the `idb` library in IndexedDB, utilizing independent object stores for resources, synchronization metadata, and historical deletion records. IndexedDB provides persistent storage with ACID transactions within a single device, making it suitable for maintaining local replicas. To avoid blocking the main thread during component mounting phases, database initialization is deferred within a

`requestIdleCallback` loop, with sequential safety enforced via lazy initialization guards on all database accessors.

Each browser context constitutes a device in the sense of Definition 4: it maintains an independent local history, its own Lamport counter, and crosses synchronization boundaries independently. The server is a Node.js HTTP service that maintains persistent current state for all resources and broadcasts data plane updates to connected devices in near-real-time. As established in Section 2.5, *S* is a shared communication and persistence layer. It does not maintain per-device state and has no knowledge of which devices have performed which local operations. Protocol implementation spans both client and server, realizing the algorithms from Section 3. The client side consists of approximately 800 lines of TypeScript across four key modules: `db.ts` (local replica management, multi-store schema orchestration, and lazy-loading boundaries), `conversationStateManager.ts` (operation execution and component state interaction), `conversationSync.ts` (the protocol coordinator decoupled from payload lifecycle cleanup paths), and `browserMigrations.ts` `backgroundSync.ts` (network synchronization and schema version state matching).

II.2 Data Structures

Each conversation resource *r* is represented as:

```
interface Conversation {
  id: string;           // Unique identifier
  title: string;        // Display name
  messages: Message[];  // Content (data plane)
  // Protocol fields -- load-bearing
  lamport: number;      // Lamport timestamp of last update
  deviceId: string;     // Device that last updated
  isDeleted: boolean;    // Current deletion state
  deletedAtLamport?: number; // Lamport timestamp of deletion
                          // Preserved through upserts (Invariant 7)
  deletedBy?: string;   // Device that issued deletion
  // Display fields -- non-protocol
  createdAt: Date;      // Human-readable creation time
  lastModified: Date;   // Human-readable last modification time
}

interface Message {
  id: string;
  role: 'user' | 'assistant';
  content: string;
  lamport: number;      // Lamport timestamp of this message
}
```

Physical timestamp fields (`createdAt`, `lastModified`) are retained for human-readable display only. They are not used in any protocol decision. All ordering decisions use the `lamport` field exclusively. The `deletedAtLamport` field requires careful handling at the server: it is set when a DELETE is received and is never cleared by subsequent data plane updates (Invariant 7). This persistence is what enables correct participation decisions at synchronization boundaries even after a resource has been restored by an upsert. The Lamport counter for each device is stored persistently (in a dedicated IndexedDB store) separate from conversation state. This ensures the counter survives application restarts and page refreshes. A device that restarts without its Lamport counter would be unable to correctly establish happens-before ordering with prior operations.

II.3 Local Operation Execution

Operations execute immediately on the local IndexedDB replica:

```

async createConversation(
  provider: string,
  model: string
): Promise<Conversation> {
  const lamport = await this.incrementLamport();
  const conversation: Conversation = {
    id: generateId(),
    title: 'New Conversation',
    messages: [],
    lamport,
    deviceId: this.deviceId,
    isDeleted: false,
    createdAt: new Date(),
    lastModified: new Date(),
    provider,
    currentModel: model
  };
  await this.db.conversations.put(conversation);
  return conversation;
}

async deleteConversation(id: string): Promise<void> {
  const conv = await this.db.conversations.get(id);
  if (!conv) return;
  const lamport = await this.incrementLamport();
  /* NB: the record is updated in place, not removed from IndexedDB.
   * Invariant 5 requires the deletion record to persist in local storage
   * so the conversationUpdated listener can distinguish this resource
   * (deleted: isDeleted=true present) from an unknown resource (no record).
   * Removing the record here would collapse Invariants 5 and 6 to the
   * same null check, making them jointly unenforceable. */
  conv.isDeleted = true;
  conv.deletedAtLamport = lamport;
  conv.deletedBy = this.deviceId;
  await this.db.conversations.put(conv);
}

async addMessage(
  conversationId: string,
  message: Omit<Message, 'lamport'>
): Promise<void> {
  const conv = await this.db.conversations.get(conversationId);
  // Invariant 5: Local Deletion Finality
  if (!conv || conv.isDeleted) {
    return;
    // Silently discard
  }
  const lamport = await this.incrementLamport();
  const stampedMessage: Message = { ...message, lamport };
  conv.messages.push(stampedMessage);
  conv.lamport = lamport;
  conv.deviceId = this.deviceId;
  conv.lastModified = new Date();
  await this.db.conversations.put(conv);
  await this.syncToServer(conv);
}

private async incrementLamport(): Promise<number> {
  const current = await this.db.lamport.get('counter') ?? 0;
  const next = current + 1;
  await this.db.lamport.put(next, 'counter');
  return next;
}

```

The silently discarded path in `addMessage` directly implements Invariant 5. No error is thrown. No state is modified. The update is simply not applied and this applies equally to any data plane operation targeting a locally deleted resource.

II.4 Data Plane Synchronization

Data plane updates propagate immediately to S:

```
async syncToServer(conversation: Conversation): Promise<void> {
  try {
    const response = await fetch('http://localhost:4001/pushChats', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ chats: [conversation] })
    });
    if (!response.ok) {
      this.markDirty(conversation.id);
    }
  } catch (error) {
    this.markDirty(conversation.id);
  }
}
```

The server applies Lamport-ordered last-write-wins merging with upsert semantics:

```
function addOrUpdateChats(newChats, serverLamport) {
  const chats = readChats();
  const byId = Object.fromEntries(chats.map(c => [c.id, c]));
  let updatedLamport = serverLamport;
  for (const chat of newChats) {
    // Advance server Lamport counter
    updatedLamport = Math.max(updatedLamport, chat.lamport) + 1;
    const existing = byId[chat.id];

    if (!existing) {
      byId[chat.id] = chat;
      broadcastToConnectedDevices(chat);
      continue;
    }

    if (existing.isDeleted) {
      // Upsert: restore resource for third-party device propagation.
      existing.isDeleted = false;
    }

    if (chat.lamport > existing.lamport) {
      existing.messages = chat.messages;
      existing.lamport = chat.lamport;
      existing.deviceId = chat.deviceId;
      existing.lastModified = chat.lastModified;
      byId[chat.id] = existing;
      broadcastToConnectedDevices(existing);
    }
  }

  writeChats(Object.values(byId));
  writeLamport(updatedLamport);
}
```

The invariant on `deletedAtLamport` is load-bearing (Invariant 7). The upsert restores the resource so connected third-party devices receive the update. But the deletion record is preserved so that devices can make correct participation decisions at their next synchronization boundary. A resource that has been upserted will have `isDeleted=false` and `deletedAtLamport` set. This combination signals to reconciling devices that a deletion occurred and that they should check their causal participation.

The client registers a listener for real-time broadcasts, implementing the receive handler of Section 3.2 and enforcing Invariants 5 and 6:

```
// Real-time listener for server-pushed data plane broadcasts
this.socket.on('conversationUpdated', async (broadcast: Conversation) => {
  const localLamport = await this.db.lamport.get('counter') ?? 0;
  await this.db.lamport.put(Math.max(localLamport, broadcast.lamport) + 1, 'counter');

  const localConv = await this.db.conversations.get(broadcast.id);

  // Invariant 6: Data Plane Non-Origination
  if (!localConv) {
    this.signalBoundaryAvailable();
    return;
  }

  // Invariant 5: Local Deletion Finality
  if (localConv.isDeleted) {
    return;
  }

  if (broadcast.lamport > localConv.lamport) {
    await this.db.conversations.put(broadcast);
  }
});
```

The guard on `localConv` directly implements Invariant 6: a broadcast for a resource the client has never seen is not applied to local state. The receive handler instead calls `signalBoundaryAvailable()`, which notifies the application layer that a synchronization boundary is ready. A badge implementation emits a UI notification and waits for the user to initiate a boundary crossing; an auto-update implementation triggers `onSynchronizationBoundary()` immediately. Both policies are compliant as the distinction is application-layer scheduling, not protocol correctness. In either case, acquisition of the unknown resource proceeds through Case 3 of the reconciliation algorithm at the next boundary, where S supplies the authoritative content. A data plane message may signal topology but may not originate it.

II.5 Control Plane Reconciliation

Control plane DELETE operations are pushed to S immediately upon local execution:

```
async pushDeletion(conversation: Conversation): Promise<void> {
  await fetch(
    `http://localhost:4001/deleteChat/${conversation.id}`,
    {
      method: 'DELETE',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        deletedAtLamport: conversation.deletedAtLamport,
        deletedBy: conversation.deviceId
      })
    }
  );
}
```

At synchronization boundaries the client executes:

```
async onSynchronizationBoundary(): Promise<void> {
  // Step 1: Advance Lamport clock from server
  const serverLamport = await this.fetchServerLamport();
  const localLamport = await this.db.lamport.get('counter') ?? 0;
  const newLamport = Math.max(localLamport, serverLamport) + 1;
  await this.db.lamport.put(newLamport, 'counter');

  // Step 2: Fetch current server state
  const response = await fetch('http://localhost:4001/fetchChats');
  const serverConversations: Conversation[] = await response.json();

  // Step 3: Reconcile each resource
  for (const serverConv of serverConversations) {
    const localConv = await this.db.conversations.get(serverConv.id);

    if (localConv?.isDeleted) {
      await this.pushDeletion(localConv);
      continue;
    }

    if (serverConv.deletedAtLamport !== undefined) {
      const τ_local = { l: localConv?.lamport ?? -1, d: localConv?.deviceId ?? '' };
      const τ_delete = { l: serverConv.deletedAtLamport, d: serverConv.deletedBy ?? '' };
      const dominated = τ_local.l > τ_delete.l || (τ_local.l === τ_delete.l && τ_local.d
> τ_delete.d);
      const participated = localConv !== undefined && dominated;

      if (!participated) {
        await this.db.conversations.delete(serverConv.id);
      } else {
        await this.syncToServer(localConv);
      }
      continue;
    }

    if (!localConv || serverConv.lamport > localConv.lamport) {
      await this.db.conversations.put(serverConv);
    }
  }

  // Step 4: Push local-only conversations to server.
  const localOnly = await this.db.conversations
    .filter(c => !serverConversations.find(s => s.id === c.id))
    .toArray();
  if (localOnly.length > 0) {
    await fetch('http://localhost:4001/pushChats', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ chats: localOnly })
    });
  }

  this.lastBoundaryLamport = newLamport;
}
```

Case 1 is evaluated first and unconditionally, directly implementing Theorem 4 (Local Deletion Stability). The continue statement ensures no locally deleted resource proceeds to Case 2 or Case 3. The participated condition in Case 2 directly implements Definition 7 (Causal Participation) via full lexicographic Lamport tuple comparison. The local device must strictly dominate the deletion record: either its logical counter exceeds the deletion's counter, or, at equal counters, its device identifier is lexicographically greater. Equal counters with a non-dominant device identifier default to deleted, ensuring deterministic resolution of perfectly concurrent operations in favor of the destructive control plane action. The default to deleted behavior for new devices falls out naturally: a device with no local record of a resource (localConv is undefined) cannot have participated and defaults to deleted.

II.6 Engineering Considerations

Lamport counter persistence. The Lamport counter must survive application restarts and page refreshes. Storing it in a dedicated IndexedDB store separate from conversation state ensures it is not accidentally cleared when conversations are modified. The counter must also be advanced correctly on server state fetch: receiving the server's current Lamport value and taking the maximum before incrementing ensures the device's counter reflects the global state of the system.

Lamport counter and device identity. Each device requires a stable, unique identifier to serve as the tie-breaker in Lamport timestamp ordering. In the proof of concept this is generated on first launch and stored in localStorage. A production implementation should use a more robust identity mechanism.

Synchronization boundary definition. The proof of concept treats application initialization as a synchronization boundary. This is one valid choice among many. Production implementations might define additional synchronization boundaries at explicit user actions, periodic intervals, or application-defined state transitions. The protocol is agnostic to this choice provided synchronization boundaries occur with sufficient frequency to bound the staleness of control plane state.

Dirty state tracking. When a data plane synchronization fails, conversations are marked dirty via a separate IndexedDB. On the next synchronization boundary, they're pushed to S before reconciliation fetch, ensuring local data plane state is reflected in S before reconciliation decisions are made.

Concurrent local contexts. Multiple browser tabs on the same device share a single IndexedDB instance, including the Lamport counter store. IndexedDB transactions serialize access to the counter, preventing duplicate Lamport values within a single device. However, the reconciliation algorithm assumes a single active context per device at each synchronization boundary. Production implementations should define explicit device boundary semantics for multi-tab scenarios.

Deletion record preservation. The most operationally critical implementation detail, formalized as Invariant 7, is that the server must never clear deletedAtLamport when applying a data plane upsert. This field is the mechanism by which reconciling devices learn that a deletion occurred and determine their causal participation. Its accidental omission would cause devices to treat upserted resources as never-deleted, preventing correct participation decisions.

Local deletion record retention. deleteConversation calls db.conversations.put(conv) rather than db.conversations.delete(conv.id). This is a protocol requirement, not an implementation convenience: the deletion record must remain in IndexedDB to allow the conversationUpdated listener to distinguish a deleted resource from an unknown one. The two cases produce different responses (silent discard versus boundary availability signal) but collapse to the same missing-record result if the deletion record is removed. A local deletion record may be purged only when a synchronization boundary reveals the resource absent from S's resource list, indicating that server-side garbage collection is complete and no future broadcasts for that resource id will arrive.

II.7 Code Availability

The Polyglot proof-of-concept implementation is available under the MIT license. The repository is under active development. A full reference implementation conforming to the formal model of Section 2 is in progress and will be made available at the same location.

Appendix III: Evaluation

We evaluate the eventual coherence protocol along two dimensions: correctness under the conflict scenarios its formal model addresses, and a preliminary characterization of performance properties. We do not claim benchmark superiority over alternative systems. The proof-of-concept nature of the Polyglot implementation precludes rigorous comparative evaluation. What we demonstrate is that the protocol behaves correctly under its target scenarios and that its performance characteristics are consistent with the complexity analysis of Section 3.8.

III.1 Experimental Setup

Our experiments use three simulated devices, implemented as separate browser contexts sharing no local state, communicating with a single server instance. A fourth context is used for new device experiments. Each browser context maintains an independent IndexedDB instance, its own Lamport counter, and crosses synchronization boundaries independently. Synchronization boundaries are triggered manually to allow precise control over the sequencing of conflict scenarios. We report Lamport timestamps alongside wall clock times where relevant. Lamport timestamps are the load-bearing ordering mechanism in the protocol. Wall clock times are provided for readability only.

All experiments run on a single machine with an Intel Core i7 processor and 16GB RAM. The client runs in Chrome 122 and the server runs Node.js 20. Each experiment is repeated 10 times and we report median values.

III.2 Correctness Evaluation

We validate the protocol’s correctness properties through systematic testing of scenarios addressed by the formal model. Each experiment targets one or more theorems or invariants from Section 4.

Experiment 1: Core scenario — delete, update, and third-party propagation. Device A deletes conversation C at Lamport timestamp $L_d=50$. A pushes the deletion to the server immediately. Device B, operating within its current synchronization epoch and unaware of the deletion, adds a message to C. B’s Lamport clock, advanced by prior activity on other resources, is at 80. B’s update carries Lamport timestamp $L_u=81$. The server receives B’s update for the deleted C, applies the upsert (restoring C, preserving $\text{deletedAtLamport}=50$), and broadcasts the restored C to Device C. Both A and B then cross synchronization boundaries.

Result. Device C receives B’s update via the upsert broadcast. Device B crosses its synchronization boundary; $\tau_{\text{local}}=(81, B) > \tau_{\text{delete}}=(50, A)$, so B retains C through causal participation. Device A crosses its synchronization boundary; Case 1 triggers (A has C locally deleted), A re-asserts the deletion to the server, and A maintains C as deleted. This confirms Theorem 2 and validates the upsert propagation mechanism: third-party devices receive updates from active participants regardless of deletion state.

Experiment 2: Local Deletion Finality under upsert pressure. Device A deletes conversation C at Lamport timestamp $L_d=50$. Device B updates C repeatedly at Lamport timestamps 60, 70, and 80, triggering three successive upserts at the server and three restoration broadcasts to A.

Result. A discards all three restoration broadcasts under Invariant 5. A's local state has C as deleted throughout. At A's next synchronization boundary, Case 1 triggers, A re-asserts the deletion, and the server marks C as deleted again. This confirms Theorem 4: A's deletion is stable across all synchronization boundary crossings and cannot be overridden by remote data plane operations regardless of their volume or Lamport values.

Experiment 3: Default to deleted for new device. Device A deletes conversation C at Lamport timestamp $L_d=50$. Device B updates C, triggering a upsert at the server (C restored, $\text{deletedAtLamport}=50$ preserved). New Device D connects and crosses its first synchronization boundary. D has no local record of C and no causal participation.

Result. D's reconciliation algorithm reaches Case 2 for C: $\text{deletedAtLamport}=50$ is set and r_local is null, so the participation check fails (no strict lexicographic dominance possible with no local state). D defaults to deleted and does not acquire C. This confirms the default-to-deleted rule: a device with no causal participation accepts the deletion regardless of the server's current restoration state.

Experiment 4: Lamport participation threshold, including equal-counter tie-break. Three sub-experiments establish the boundary of the participation check, including the tie-break behavior introduced by the full lexicographic comparison.

- Sub-experiment 4a (participation confirmed). Device B has been active across multiple conversations, advancing B's Lamport clock to 120. Device A deletes conversation C at Lamport timestamp $L_d=100$. B updates C at Lamport timestamp $L_u=121$. At B's synchronization boundary: $\tau_local=(121, B) > \tau_delete=(100, A)$. B retains C.

- Sub-experiment 4b (non-participation confirmed). Device B has been inactive. B's Lamport clock is at 40. Device A deletes C at Lamport timestamp $L_d=100$. B updates C at Lamport timestamp $L_u=41$. At B's synchronization boundary: $\tau_local=(41, B) \not> \tau_delete=(100, A)$. B accepts the deletion.

- Sub-experiment 4c (equal-counter tie-break, defaults to deleted). Device B updates C with Lamport counter $l=75$. Device A deletes C with Lamport counter $l=75$. At B's synchronization boundary: $l_local = l_delete = 75$. Since $B < A$ lexicographically ($B \leq A$), $\tau_local \not> \tau_delete$, and B defaults to deleted. This confirms deterministic resolution of perfectly concurrent operations in favor of the destructive control plane operation, without cross-device coordination.

Result. The full lexicographic Lamport participation check correctly distinguishes causal participation from non-participation and resolves equal-counter ties deterministically. This validates Definition 7 as implemented.

Experiment 5: Control plane boundary delay. Device A creates conversation X within its synchronization epoch. Device B creates conversation Y within its synchronization epoch. Neither has crossed a synchronization boundary. Both then cross synchronization boundaries.

Result. Both X and Y appear on both devices after synchronization boundary crossings. New resources created within a synchronization epoch propagate correctly at the boundary through Step 4 of the reconciliation algorithm. This confirms property (1) of the Problem Statement in Section 2.7: devices perform operations continuously within epochs without coordination. These five experiments validate the protocol's correctness properties: per-device causal validity (Theorem 1), convergence to causally valid states (Theorem 2), update preservation (Theorem 3), local deletion stability (Theorem 4), and the default-to-deleted rule.

III.3 Performance Characteristics

We report preliminary performance observations consistent with the complexity analysis of Section 3.8. These are not rigorous benchmarks.

Lamport counter overhead. Lamport counter increment and persistence (IndexedDB write) adds under 1ms per operation. This overhead is negligible relative to network round-trip time and does not materially affect data plane latency.

Latency. Data plane updates complete locally in under 5ms and propagate to the server in 100 to 150ms over a local network connection, dominated by network round-trip time. Control plane reconciliation at a synchronization boundary takes 50 to 200ms depending on the number of conversations, consistent with the $O(|R|)$ complexity of Section 3.8.

Throughput. The proof of concept handles in excess of 100 data plane updates per second per device without coordination bottlenecks. Lamport counter operations add no measurable throughput constraint.

Storage. In a test dataset of 137 conversations with between 5 and 50 messages each, total storage per device was 2.3MB. The lamport and deletedAtLamport fields added under 2KB of overhead across the dataset. Deletion records added less than 1KB of additional metadata.

III.4 Preliminary Comparison with Alternative Approaches

We implemented simplified versions of three alternative approaches within the same proof-of-concept harness to characterize behavioral differences. These are not rigorous comparative benchmarks. The alternative implementations are not optimized and the workload is not designed to stress them systematically.

Last-Write-Wins. We implemented a system in which DELETE is treated as a timestamped write operation competing with UPDATE on equal terms, using Lamport timestamps for ordering.

Result. In this model, deletion and update compete under the same ordering. If B's update carries a higher Lamport timestamp than A's deletion, B's update wins globally: C is restored on all devices including A. A, which explicitly deleted C, sees C restored. There is no equivalent of Invariant 5: the deleting device loses its deletion when outpaced by a more active retaining device. Users of the proof of concept reported this as confusing and contrary to expectation.

CRDT (Observed-Remove Set). We implemented deletion using an observed-remove set. Resources accumulate deletion tombstones that are never garbage collected without external coordination.

Result. After 1000 operations including 100 deletions, storage grew to 4.1MB compared to 2.3MB for eventual coherence, with tombstones comprising 45% of total storage. The deletedAtLamport field in eventual coherence serves a similar bookkeeping function but is a single integer per deleted resource, eligible for garbage collection once all devices have reconciled. Eventual coherence achieves true deletion without unbounded storage growth.

Coordination-based (Two-Phase Commit). We implemented a two-phase commit protocol requiring acknowledgment from all participating devices before a deletion is finalized, with Lamport-ordered acknowledgments.

Result. When any device was slow to cross a synchronization boundary, deletions blocked indefinitely. Average deletion latency was 800ms compared to under 5ms for local deletion in eventual coherence. The system violated availability during control plane boundary delays, consistent with the CAP theorem analysis of Section 2.7.

III.5 Observational Notes

During informal use of the Polyglot proof of concept across multiple personal devices over approximately one month, no instances of data loss were observed. The synchronization boundary behavior matched the protocol’s stated properties: resources modified within a synchronization epoch were retained after boundary crossings when causal participation was established, and resources not modified were correctly updated or deleted based on server state.

One interaction pattern confirmed Theorem 4 in practice. A conversation deleted on one device remained deleted on that device across multiple synchronization boundary crossings, even while another device continued adding messages to the same conversation. The retaining device continued to function normally. Third-party devices received updates from the retaining device through the upsert mechanism. The deleting device’s state was unaffected throughout.

The default-to-deleted behavior for new devices was also observed. A new browser context connecting for the first time did not acquire conversations carrying deletion records, regardless of whether those conversations had been restored by another device’s data plane updates.

III.6 Limitations

Scale and Causal Horizons. The protocol’s reliance on scalar Lamport timestamps rather than vector clocks introduces a distinct causal horizon boundary for isolated operations. If a device executes a local update while its local logical clock is lagged relative to a remote deletion event ($l_{\text{local}} \leq l_{\text{delete}}$), the reconciliation algorithm will mathematically categorize the update as historically superseded. This represents the formal limits of the protocol’s correctness framework: historical validity is guaranteed only for devices operating within the active causal horizon of the cluster. A device that fails to advance its local Lamport counter relative to the cluster’s control plane mutations effectively drops out of the causal sequence, and its concurrent mutations are treated as causally inert upon reconciliation.

Network conditions. All experiments used local network conditions with round-trip times of 100 to 500ms. Wide-area network conditions with higher latency or more complex partition patterns may surface behaviors not observed in these experiments, though the protocol’s correctness properties are not dependent on network latency bounds.

Workload generality. The workload is representative of single-author conversation management but may not generalize to domains with substantially higher write contention, more frequent control plane operations, or different ratios of data plane to control plane activity. In particular, the model explicitly excludes concurrent multi-author data plane operations on the same resource. In the LLM conversation domain, this means the protocol does not handle concurrent turns from different devices targeting the same conversation: the Lamport-ordered last-write-wins merge at the resource level would discard one device’s turn, violating Property 6. Furthermore, even if turn ordering were resolved technically, the semantic validity of the resulting transcript would be compromised, since each LLM-generated response is conditioned on the context visible at the time of generation. A merged transcript produced by interleaving turns from distinct causal contexts does not correspond to any conversation that any device actually had. Multi-author and concurrent-turn scenarios are therefore outside the scope of this model and are left to subsequent work.

Comparative evaluation. The alternative implementations evaluated in Section 6.4 are not optimized and the comparison is qualitative rather than quantitative. Rigorous comparative benchmarking against production-quality implementations of alternative approaches is left to future work.

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Notes for Author Review

- [12] and [39]: Same paper (Ahamad et al. 1995). Consolidate to a single citation number or confirm dual-citation is intentional.
- [15] and [88]: Same paper (Shapiro et al. SSS 2011). Consolidate or confirm intentional.
- [20]: INRIA TR 7686 (July 2011), distinct from the SSS 2011 conference paper [15] and the January 2011 TR [14]. Three Shapiro et al. works cited; confirm all three are intentionally distinct.
- [63]: The Byzantine Generals paper is an unusual choice for the quorum systems context. Thomas, R. H. 1979. A majority consensus approach to concurrency control. *ACM Trans. Database Syst.* 4, 2, 180–209 — may be the intended citation. Author should confirm.
- [65]: Placeholder (MAAN) is weak for client-server synchronization protocols. Author should supply the intended citation.
- [85]: Same paper as [29]. Author should confirm whether a distinct idempotency/retry reference was intended, or consolidate.
- [90]: Burns et al. 2016 ("Borg, Omega, and Kubernetes")